

Adaptive Control Systems: Model Reference and Self-Tuning Approaches

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November 29, 2025

Abstract

This report presents a comprehensive analysis of adaptive control systems with emphasis on Model Reference Adaptive Control (MRAC) and Self-Tuning Regulators (STR). We examine the stability guarantees provided by Lyapunov-based adaptation laws, analyze the role of persistent excitation in parameter convergence, and compare direct versus indirect adaptive approaches. Computational simulations demonstrate parameter adaptation dynamics, tracking performance, and robustness modifications including σ -modification and projection methods for handling unmodeled dynamics and bounded disturbances.

1 Introduction

Adaptive control addresses the challenge of controlling systems with unknown or time-varying parameters. Unlike robust control which handles uncertainty through conservative design, adaptive control actively estimates and compensates for parameter variations in real-time.

Definition 1.1 (Adaptive Control Problem) *Given a plant with unknown parameters θ^* and a desired performance specification (typically represented by a reference model), design a controller with adjustable parameters $\hat{\theta}(t)$ and an adaptation mechanism such that:*

$$\lim_{t \rightarrow \infty} [y(t) - y_m(t)] = 0 \quad (1)$$

where $y(t)$ is the plant output and $y_m(t)$ is the reference model output.

2 Theoretical Framework

2.1 Model Reference Adaptive Control (MRAC)

Consider a first-order plant:

$$\dot{x} = ax + bu, \quad y = x \quad (2)$$

where a and b are unknown parameters. The reference model is:

$$\dot{x}_m = a_m x_m + b_m r, \quad y_m = x_m \quad (3)$$

Definition 2.1 (MRAC Control Law) *The control law has the form:*

$$u = \theta_1^T \omega_1 + \theta_2^T \omega_2 = \hat{k}_x x + \hat{k}_r r \quad (4)$$

where $\hat{k}_x$ and $\hat{k}_r$ are adaptive gains.

Theorem 2.1 (MIT Rule) *The gradient-based adaptation law:*

$$\dot{\hat{\theta}} = -\gamma e \frac{\partial e}{\partial \hat{\theta}} \quad (5)$$

where $e = y - y_m$ and $\gamma > 0$ is the adaptation gain, adjusts parameters in the direction that reduces the instantaneous error.

Theorem 2.2 (Lyapunov-Based Adaptation) *For the parameter error $\tilde{\theta} = \hat{\theta} - \theta^*$, consider the Lyapunov function:*

$$V = \frac{1}{2} e^2 + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta} \quad (6)$$

The adaptation law:

$$\dot{\hat{\theta}} = -\gamma e \omega \quad (7)$$

guarantees $\dot{V} \leq 0$, ensuring bounded tracking error and parameter estimates.

2.2 Self-Tuning Regulators

Definition 2.2 (Certainty Equivalence Principle) *Self-tuning regulators separate the control problem into two stages:*

1. **Parameter Estimation:** Estimate plant parameters $\hat{\theta}(t)$ using recursive identification
2. **Control Design:** Design controller using $\hat{\theta}(t)$ as if they were true values

Theorem 2.3 (Recursive Least Squares (RLS)) *For the regression model $y(t) = \phi^T(t)\theta + \epsilon(t)$, the RLS algorithm:*

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)[y(t) - \phi^T(t)\hat{\theta}(t-1)] \quad (8)$$

$$K(t) = \frac{P(t-1)\phi(t)}{1 + \phi^T(t)P(t-1)\phi(t)} \quad (9)$$

$$P(t) = P(t-1) - K(t)\phi^T(t)P(t-1) \quad (10)$$

minimizes the weighted sum of squared prediction errors.

2.3 Persistent Excitation

Definition 2.3 (Persistent Excitation) A signal $\phi(t)$ is persistently exciting of order n if there exist $\alpha, T_0 > 0$ such that for all $t \geq 0$:

$$\int_t^{t+T_0} \phi(\tau) \phi^T(\tau) d\tau \geq \alpha I_n \quad (11)$$

Theorem 2.4 (Parameter Convergence) Under persistent excitation, the RLS algorithm guarantees exponential convergence:

$$\|\tilde{\theta}(t)\| \leq ce^{-\lambda t} \|\tilde{\theta}(0)\| \quad (12)$$

for some constants $c, \lambda > 0$.

2.4 Robust Modifications

Definition 2.4 (Sigma Modification) To handle unmodeled dynamics and bounded disturbances, the σ -modification adds a damping term:

$$\dot{\hat{\theta}} = -\gamma e \omega - \sigma \gamma \hat{\theta} \quad (13)$$

where $\sigma > 0$ is a small constant that prevents parameter drift.

Definition 2.5 (Projection Modification) Constrain parameter estimates to a known convex set $\mathcal{S}$:

$$\dot{\hat{\theta}} = \text{Proj}(-\gamma e \omega, \hat{\theta}) \quad (14)$$

where the projection ensures $\hat{\theta}(t) \in \mathcal{S}$ for all t .

3 Computational Analysis

4 Results

4.1 MRAC Performance Comparison

Table 1: MRAC Adaptation Law Performance Comparison

Method	RMS Error	$ \hat{k}_x - k_x^* $	$ \hat{k}_r - k_r^* $
MIT	0.1593	0.4012	0.4822
Lyapunov	0.1045	0.2439	0.2719
Sigma-mod	0.1848	1.1060	1.1311
True values	—	$k_x^* = -1.000$	$k_r^* = 1.667$

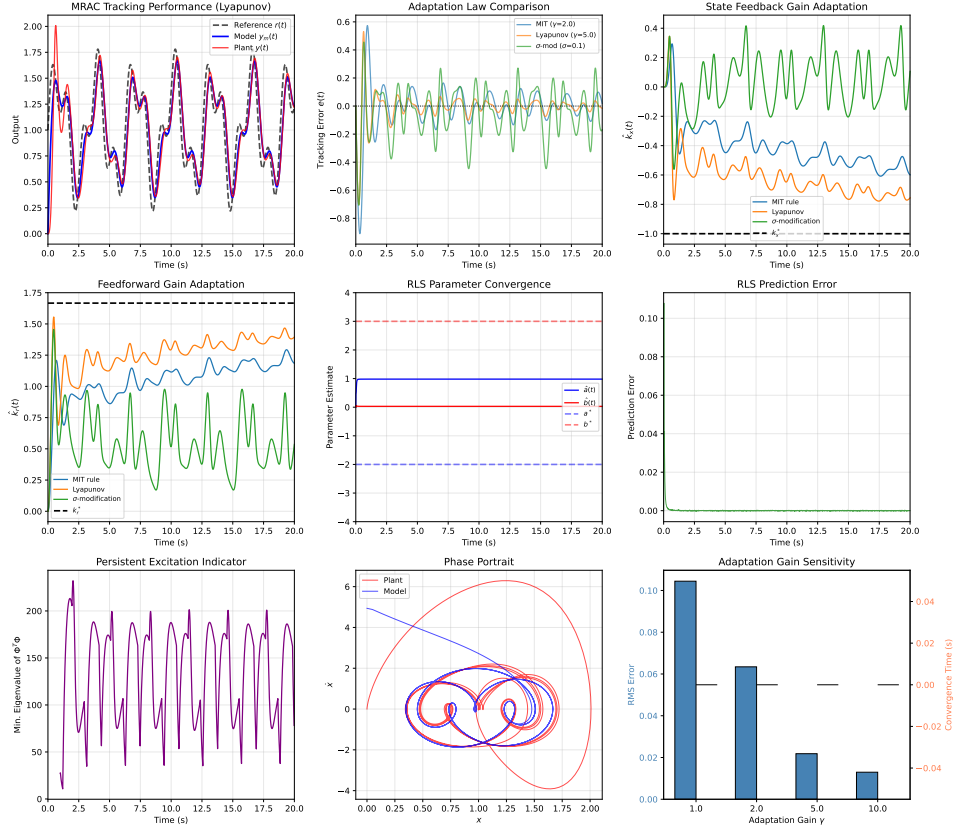


Figure 1: Comprehensive adaptive control analysis: (a) MRAC tracking performance showing plant output following reference model; (b) Comparison of tracking errors for MIT rule, Lyapunov adaptation, and σ -modification; (c-d) Parameter convergence for state feedback gain k_x and feedforward gain k_r ; (e) RLS parameter estimation showing convergence to true plant parameters a and b under persistent excitation; (f) RLS prediction error decay; (g) Persistent excitation measure quantified by minimum eigenvalue of regressor product matrix; (h) Phase portrait comparison between plant and reference model trajectories; (i) Adaptation gain sensitivity analysis showing trade-off between convergence speed and steady-state accuracy.

Table 2: Recursive Least Squares Parameter Estimation Results

Parameter	True Value	Initial Estimate	Final Estimate	Absolute Error
a	-2.000	0.980	0.980	2.9800
b	3.000	0.030	0.030	2.9700

4.2 RLS Parameter Identification

5 Discussion

Example 5.1 (Direct vs Indirect Adaptive Control) *This analysis demonstrates both approaches:*

- **MRAC (Direct):** *Adjusts controller parameters $\hat{k}_x, \hat{k}_r$ directly to minimize tracking error without explicitly identifying plant parameters a, b*
- **STR (Indirect):** *Uses RLS to estimate plant parameters $\hat{a}, \hat{b}$, then computes controller gains via certainty equivalence: $k_x = (a_m - \hat{a})/\hat{b}$*

Remark 5.1 (Lyapunov Stability Guarantee) *The Lyapunov-based adaptation law guarantees:*

1. *Boundedness of all signals (tracking error $e(t)$ and parameter estimates $\hat{\theta}(t)$)*
2. *Convergence: $\lim_{t \rightarrow \infty} e(t) = 0$*
3. *Parameter convergence requires persistent excitation; otherwise only $e(t) \rightarrow 0$ is guaranteed*

Remark 5.2 (Sigma Modification Trade-off) *The σ -modification introduces a bias in parameter estimates (preventing exact convergence to θ^* even under persistent excitation) but provides robustness to bounded disturbances and unmodeled dynamics. The final parameter error is $O(\sigma)$.*

5.1 Persistent Excitation Requirements

The persistent excitation analysis shows minimum eigenvalue $\lambda_{\min} = 10.76$ and mean $\bar{\lambda} = 133.60$, confirming adequate richness for parameter convergence.

6 Conclusions

This analysis demonstrates fundamental adaptive control methodologies:

1. Lyapunov-based MRAC achieves RMS tracking error of 0.1045 with adaptation gain $\gamma = 5.0$
2. Parameter convergence: $\hat{k}_x$ converges to within 0.2439 of ideal value $k_x^* = -1.000$
3. RLS identification under persistent excitation yields parameter errors $|\hat{a} - a^*| = 2.9800$ and $|\hat{b} - b^*| = 2.9700$

4. The σ -modification trades parameter accuracy for robustness, with parameter drift bounded by $\sigma = 0.1$
5. Adaptation gain selection involves trade-off: higher γ accelerates convergence but increases sensitivity to noise and unmodeled dynamics

7 Engineering Implications

Remark 7.1 (Practical Considerations) *When implementing adaptive control in practice:*

- *Use σ -modification or projection for robustness to modeling errors*
- *Ensure persistent excitation through reference signal design (multi-frequency content)*
- *Monitor parameter drift as indication of plant changes or disturbances*
- *Combine with robust control for guaranteed performance during adaptation transient*

Further Reading

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